

SUPPLEMENTARY MATERIAL S1

Candidate readings of the Xiongnu record

Thirty-four readings of Chinese-transcribed Xiongnu names and titles as Turkic words, with the historical parallels that suggested each one. Supplied for the record; not offered as results.

Aziz M. Uluğ

Supplement to *Reading steppe names: what a transcription channel can and cannot recover from the Chinese record of the Xiongnu*

S1 · author_proposals.csv · compiled 2026

S1.1 What this file is

Section 9 of the paper sets out what a proposed reading of a Xiongnu name has to survive before it can be called a finding: the sense has to be attested in a source independent of the comparison, the morphology has to be licensed by a suffix with the right distribution, and any objection that can be tested has to be tested. One reading in this project meets that standard. The thirty-three others do not.

They are here anyway. They were produced over the course of the work, they are the readings the author actually finds attractive, and suppressing them would be its own kind of dishonesty — a reader who has seen 稽粥 written out as *kei çuk* will hear Turkish *küçük* whether or not this file exists. What can be done is to write them down, date them, label them, and put next to each one the reason it occurred to the author, so that the reasoning is inspectable rather than implied.

Nothing in the paper depends on any row below. The measurements in §§5–8 — the channel accuracies, the period-gap decomposition, the initial-consonant probe, the coda-spelling count — were made without reference to these readings and do not change if every one of them is rejected.

S1.2 Why they are not in the paper

Section 6.3 measures exactly the operation that produced this file — taking a reconstructed reading and looking it up in a Turkic dictionary — against a null model built by resampling Later Han syllables into pseudo-names. Across seven lexicons, from modern Turkish to the *Codex Cumanicus*, between none and three names in thirty-six clear $p \leq 0.05$. At a one-in-twenty threshold, thirty-six coin flips give the same.

Doing the matching by hand does not improve on the algorithm. It makes it worse, and in a way that is easy to state: a person cannot help preferring the reading that means something, and that preference is precisely the selection effect the null exists to quantify. The algorithm at least fails honestly. A paper that measured the unreliability of this operation and then presented thirty-four of its outputs as discoveries would be incoherent.

So the two things are kept apart. The paper reports what can be measured. This file records what was imagined while measuring it. Because it is supplementary, it does not carry the paper's evidential standard and should not be read as though it did.

S1.3 How the readings were produced

Every row starts from the same two published inputs, neither of them model output. The Chinese form is as the *Shiji*, *Hanshu* and *Hou Hanshu* write it. The Later Han reading is Schuessler's, character by character. The third column simply respells that reading in Turkish orthography, so that a reader who does not read phonetic notation can see what is being claimed; it adds no information.

The fourth column is the author's. It is arrived at by hearing the third column as a Turkic word or compound, usually with a vowel changed, sometimes with a consonant treated as lost, and then looking for a historical parallel that would make such a name plausible for a steppe ruler. The fifth column gives the sense that reading would carry. The note beneath each row gives the parallel. Each entry below is set out in two parts: *Phonetics*, which walks the Later Han reading to the proposed name character by character and says which of the steps are ordinary correspondences and which are not; and *Reason*, which gives the word, its attestation where it has one, and the naming parallel that suggested it.

Three habits of reasoning recur, and it is worth naming them because they are the file's main weakness. The first is the epithet argument: later Turkic rulers are commonly named by a common noun or a natural force — Bayezid I *Yıldırım* "the thunderbolt", Alp Arslan "the lion", Mehmed IV *Avcı* "the hunter", Süleyman I *Kanunî* "the lawgiver" — so a reading that yields such a noun is treated as more likely. The second is the career argument: where the Chinese record says what a ruler did, a reading that names the deed is preferred. The third is the compound argument: a long transcription is split into two or three Turkic elements. Each of these makes a reading easier to reach, and none of them makes it more likely to be right. They multiply the search space, which is the opposite of evidence.

S1.4 The one exception, which is argued in the paper

The recurring element 若鞮 *ṛak te*, which closes the formal names of six later chanyu, is read *Inakt*, "the faithful ones". It is in the paper as §9 rather than here, because it is supported differently from everything else in this file:

- the *Hou Hanshu* glosses 若鞮 as 孝 "filial" — the meaning is recorded in the source, not inferred from a resemblance;
- inak* is attested as *fidelis* "faithful" in the *Codex Cumanicus* (Kuun 1880, p. 182), a medieval source with no connection to the Chinese record;
- the morphology is licensed: *-t* is an Old Turkic plural used with titles (*tegin* → *tegit*, *tarqan* → *tarqat*), and the competing *-tI* reading was ruled out by counting — of 31 Cuman headwords in *-tI*, almost all are past-tense verbs;
- the obvious objection to the spelling was tested and failed: if the source ended in *-t*, why did the scribe use the open character 鞮? Because 30% of stop-final source syllables in the Sanskrit corpus were written with an open character.

One objection stands and is stated in the paper: if the source word was *inakt*, the initial *ɪ-* is not written, and 若鞮 has no room for it. It appears in the tables below inside the six compounds, so that this file can be read as one list with the standard visible in it.

S1.5 Why the decoder's output is not cited below

The channel of §4 has a decoding step that will return a ranked list of readings for any string in the record. None of those rankings is quoted in this file, and the reason is worth stating, because the numbers look more informative than they are.

A character can be *reading-anchored*, meaning its emission distribution was learned for that character specifically, or it can fall back to a table keyed on the Later Han **initial consonant class alone**, discarding the vowel and the coda. Of the 39 items in the record, only five are reading-anchored throughout; 34 use backoff on at least one character and nine use it on every character. The effect is that whole groups of characters share one distribution:

LATER HAN INITIAL CLASS	CHARACTERS IN THE XIONGNU RECORD SHARING ONE EMISSION DISTRIBUTION	WHAT THE BACKOFF TABLE RETURNS FOR EVERY ONE OF THEM
velar	17 — 稽 渠 胡 湖 壺 戶 侯 姑 孤 骨 谷 韓 諧 皋 權 胸 徑	ka 0.34, ga 0.15, ku 0.08
affricate	6 — 且 粥 珠 臣 詹 郅	ca 0.18, ja 0.10, cā 0.08
dental stop	9 — 鞮 題 當 屠 撐 株 稚 脫 頓	ta 0.23, da 0.11, ti 0.10
liquid	6 — 廬 孿 祿 累 蠡 閭	ra 0.37, la 0.17, rī 0.08

TABLE S1 Because 稽 *k^hei^a* and 渠 *giə* draw on the same row, and 粥 *t^suk* and 且 *t^sa^a* on another, an affricate-plus-velar string and a velar-plus-affricate string return mirror images of one table entry rather than anything specific to those characters. Two further limits apply: the scores are normalised over a retained beam, so they are relative rankings within it and not posteriors; and the syllable inventory is Sanskrit-derived, so *q* and the front rounded vowels *ö* and *ü* cannot be produced at all — a reading containing *ü* is therefore not rejected by the decoder, it is unavailable to it. §6.2 of the paper makes the general point; this table is its mechanism.

What is cited in the entries below, where a spelling question arises, is the initial-correspondence table of step 15: direct counts of which Chinese initial was used for which source sound across the three corpora. Those are observations rather than model output, and they are what the word "permitted" means wherever it appears in this file.

S1.6 Chanyu, in order of reign

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING PROPOSED BY THE AUTHOR	SENSE PROPOSED FOR THAT READING
頭曼 Touman	do man ^c	do man	Duman	dust cloud

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING PROPOSED BY THE AUTHOR	SENSE PROPOSED FOR THAT READING
PHONETICS 頭 <i>do</i> → <i>du</i>: the initial is direct; o and u are not separable at the ~50% vowel recovery §7 reports. 曼 <i>man</i> → <i>man</i>: exact, and the coda -n is written, because -n is one of the six codas the Chinese syllable can carry. One vowel change in two syllables. The rival reading <i>tümen</i> needs ü, which the Later Han script cannot write at all. REASON <i>duman</i> / <i>tuman</i> / <i>touman</i> “smoke, fog, mist” is one word across Kāšgarī, the Codex Cumanicus and modern Turkish. The standard identification is <i>tümen</i> “ten thousand”, which is absent from the Cuman vocabulary entirely.				
稽粥 Jizhou / Jiyu	k ^h ei ^B tśuk	kei çuk	Kiçük	the younger
PHONETICS 稽 <i>k^hei</i> → <i>ki</i>: an aspirated velar is the ordinary vehicle for a source k, and the diphthong <i>ei</i> for <i>i</i> is routine. 粥 <i>tśuk</i> → <i>çük</i>: tś- is the default character class for a source affricate (37% in the Sanskrit corpus), and the coda -k is actually written — the strongest part of the reading, since a stop coda goes unwritten 30% of the time. Only ü is unavailable and surfaces as u. REASON <i>kiçig</i> / <i>kiçik</i> / <i>kiçük</i> “small, younger” is attested three times independently: Kāšgarī has <i>kiçik</i> and <i>kiçük</i> with the derivative <i>kiçiglik</i>, the <i>Kutadgu Bilig</i> has <i>kiçig oğlan erken</i> “when he was a small boy” (1823), and the Codex has <i>Kečak</i> glossed <i>paucus</i>. 稽粥 is the personal name of Laoshang chanyu, Modu's son and successor, so “the younger” fits the succession the Chinese sources record.				
軍臣 Junchen	kun džin	kun cin	Künčin	bright as the sun
PHONETICS 軍 <i>kun</i> → <i>kün</i>: initial and coda direct; ü is unwritable and surfaces as u. 臣 <i>džin</i> → <i>čin</i>: dž is the voiced partner of tś, and voicing is leaky in both directions in every corpus measured, so a source ç is reachable through it; the -n coda is written. REASON <i>kün</i> “sun, day”. The Seljuk regnal names Kaykhusraw and Kayqubad carry an association of royal solar radiance through their <i>kay</i>- element.				
詹師廬 Zhanshilu	tśam ři lia	çam ři lia	Çamşılı	dazzling, radiant
PHONETICS 詹 <i>tśam</i> → <i>çam</i>: tś- is the default affricate spelling and the -m coda is written. 師 <i>ři</i> → <i>řr</i>: the retroflex sibilant is the standard vehicle for ř. 廬 <i>lia</i> → <i>lr</i>: note that Chinese l- also carries source r- — in step 15 an r- was written l- in 12 of 14 attestations — so this syllable is genuinely ambiguous between -lr and -rl. REASON The same solar association proposed for <i>Künčin</i>, from a stem of shining or dazzling.				
兜犁湖 Xulihu / Goulihu	huo lei ga	huo lei ga	Kalık	sky, heaven
PHONETICS 兜 <i>huo</i> → <i>ka</i>, 犁 <i>lei</i> → <i>lr</i>, 湖 <i>ga</i> → <i>k</i>: three onsets, each taken from the majority correspondence, with nothing inserted and no coda discarded. The velar class covers h-, k- and g- alike, so the first and third syllables cost nothing. REASON Kāšgarī glosses <i>kalık</i> “hava, gök, sema” — air, sky, heaven. The word belongs to the same semantic field as 撐犁, the one Xiongnu term the Chinese sources gloss (天 “heaven”), so a ruler bearing it would not be isolated in the record. A second exact fit, <i>kılık</i> “disposition, manner”, is available if the sky reading is thought too convenient.				
伊稚斜 Yizhixie	?i dī ^c ja	’i di ya	İdiya	master
PHONETICS 伊 <i>?i</i> → <i>i</i>: a glottal onset is the standard way to write a vowel-initial syllable. 稚 <i>dī</i> → <i>dī</i>: retroflex d for d is routine. 斜 <i>ja</i> → <i>ya</i>: this character class emits <i>ya</i> more strongly than any other single-syllable correspondence in the record. All three steps are default. REASON <i>idi</i> “lord, master”, attested in Old Turkic. Compare Murad I's epithet <i>Hüdavendigâr</i>, “the sovereign”, a title of the same shape and force.				
烏維 Wuwei	?a wi	’a vi	Abı	one who hunts or stalks
PHONETICS 烏 <i>?a</i> → <i>a</i>: vowel-initial, direct. 維 <i>wi</i> → <i>bı</i> is the weak step: the Chinese character carries a w-, and while Turkic b- and w- alternate, the transcription reads more naturally as <i>avı</i> than <i>abı</i>. This is one of only five items in the record decoded entirely from character-specific evidence, so the mismatch cannot be attributed to a gap in the data. REASON <i>av</i> “hunt”, <i>avcı</i> “hunter”. <i>Avcı</i> names an occupation; <i>abı</i> is proposed as naming a trait. <i>Avcı</i> is also the epithet of Mehmed IV.				
且鞮侯 Qiedihou	tśa ^B te go	tśa te go	Çatıgka	one ready for the clash
PHONETICS 且 <i>tśa</i> → <i>ça</i>: ts- for a source affricate is a minority spelling but a real one, 15% in the Sanskrit corpus and 9% in Ligeti. 鞮 <i>te</i> → <i>tr</i>: direct. 侯 <i>go</i> → <i>gka</i> asks a single Chinese syllable to carry a consonant cluster it structurally cannot, so the reading adds material the transcription does not have. REASON <i>çat</i>- “to strike together, to clash”. Compare Selçuk Bey, whose name has been read as “perpetually ready for the fight”.				
狐鹿姑 Hulugu	ɣua lok ka	ğua lok ka	Kurluga	the encloser, the binder

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING PROPOSED BY THE AUTHOR	SENSE PROPOSED FOR THAT READING
PHONETICS 姑 <i>ka</i> → <i>ga</i> is well supported by the velar correspondence. The other two syllables are not: 狐 <i>yua</i> → <i>kur</i> supplies an -r that Later Han has no coda slot for, and 鹿 <i>lok</i> → <i>lu</i> discards a coda that was written. Adding an unwritable segment and dropping a written one in the same word pulls in opposite directions. REASON 狐鹿姑 captured the Han general Li Guangli and held the confederation together through a hard decade, so an epithet of binding fits the record. Compare the Göktürk ruler Kapgan Kagan, <i>kapgan</i> “the encloser”.				
壺衍鞬 Huyandi	ga jan ^B te	ga yan te	Kayanti	avalanche
PHONETICS 壺 <i>ga</i> → <i>ka</i> : given a Later Han voiced g-, the source sound is more often k than g, so the devoicing is normal rather than a liberty. 衍 <i>jan</i> → <i>yan</i> : direct, coda written. 鞬 <i>te</i> → <i>ti</i> : direct. All three steps are default or better — phonetically one of the cleaner readings here. REASON <i>kay-</i> “to slide, to slip away”. Parallel in sense to Bayezid I's <i>Yıldırım</i> : a natural force taken as a regnal epithet.				
虛間權渠 Xulüquanqu	k ^h ia lia gyan gĩa	kia lia gyan gĩa	Kalkan	shield
PHONETICS 虛 <i>k^hia</i> → <i>ka</i> , 間 <i>lia</i> → <i>l</i> , 權 <i>gyan</i> → <i>kan</i> : the first three characters match exactly, including the written -n coda, at zero minority steps. Only 渠 <i>gĩa</i> is left unaccounted for, and it may carry a separate title element. REASON <i>kalkan</i> “shield” is in Kāšgari and in the Codex Cumanicus as <i>Kalkani</i> , “scutum ejus”. A weapon or armour name is an ordinary steppe formation, and it needs no compound to be read.				
握衍胸鞬 Woyanqudi	ʔok jan ^B guo te	’ok yan guo te	Okyan Kutuğti	blessed sovereign of the arrow flank
PHONETICS 握 ʔok → <i>ok</i> is exact, with the -k coda written, and 衍 <i>jan</i> → <i>yan</i> and 鞬 <i>te</i> → <i>ti</i> are direct. The weak point is 胸 <i>guo</i> → <i>kutuğ</i> , which expands one Chinese syllable into two Turkic ones; nothing in the corpora licenses that expansion. REASON <i>ok</i> “arrow” doubles as a division of the confederation in later Turkic usage — the <i>On Ok</i> , “ten arrows”, of the Western Turks — which is what makes a territorial reading available. <i>kut</i> is “heavenly favour, fortune”.				
呼韓邪 Huhanye	ha gan ja	ha gan ya	Kaganya	kagan of the flank
PHONETICS 呼 <i>ha</i> → <i>ka</i> : an h/k substitution, which the Mongolian corpus supports strongly (a source k or q was written with an h- character 27% of the time). 韓 <i>gan</i> → <i>gan</i> : exact, coda written. 邪 <i>ja</i> → <i>ya</i> : the strongest single correspondence in the record. The written sequence <i>ha gan ya</i> becomes <i>ka-gan-ya</i> on one change. REASON 呼韓邪 commanded the left flank before becoming ruler, so the Chinese record itself supplies the office that the reading names.				
郅支 Zhizhi	tśit tśe	çit çe	Çitçi	border-maker, one who carves the boundary
PHONETICS 郅 <i>tśit</i> → <i>çit</i> : tś- is the default affricate vehicle and the -t coda is written. 支 <i>tśe</i> → <i>çi</i> : default. Both syllables use the majority correspondence and neither requires an inserted or deleted segment. REASON <i>çit</i> / <i>çiz-</i> , of marking or carving a boundary. 郅支 broke away westward and set up a rival seat, so a name of division fits the career — though it is equally possible that the career suggested the name.				

TABLE S2 The fourteen rulers whose names are not built on the recurring element 若鞬. Reign order follows the *Shiji* and *Hanshu* as reproduced in Appendix A of the paper.

S1.7 The six 若鞬 compounds

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING PROPOSED BY THE AUTHOR	SENSE PROPOSED FOR THAT READING
復株累若鞬 Fuzhulei Ruodi	buk ʈio lui ńak te	buk tio lui nyak te	Büktel Inakt	Büktel, of the Inakt
PHONETICS 復株累 <i>buk ʈio lui</i> → <i>büktel</i> : the b-, the -k coda, the dental and the liquid all fall in place across the three characters, at zero minority steps, with only the front rounded vowel unwritable. One attested word covers the whole prefix rather than an invented two-element compound. 若鞬 → <i>Inakt</i> is treated in §S1.4. REASON Kāšgari gives <i>büktel</i> as “orta boylu; yassı arkalı, oturamaklı” — of middling height and flat-backed. A name from a bodily description is a well-attested Turkic type; compare the Hungarian royal name <i>Şişman</i> .				
搜諧若鞬 Souxie Ruodi	so gei ńak te	so gei nyak te	Kemiş-Ok Inakt	Arrow of Tribute, of the Inakt

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING PROPOSED BY THE AUTHOR	SENSE PROPOSED FOR THAT READING
PHONETICS The two elements before 若鞮 are not derivable from the transcription by any correspondence in the corpora: 搜 <i>so</i> → <i>kemiş</i> would require an s- character to carry k-m-ş, and 諧 <i>gei</i> → <i>ok</i> would require a velar-initial character to yield a vowel-initial syllable. This reading rests on the sense rather than on the spelling. REASON <i>kemiş</i> of offering or casting, with <i>ok</i> “arrow” as the division element seen in the other compounds.				
車牙若鞮 Cheya Ruodi	tʃʰa ɲa ɳak te	ça nga nyak te	Çöge-Ok Inakt	Arrow of High Eminence, of the Inakt
PHONETICS 車 <i>tʃʰa</i> → <i>çöge</i> : the aspirated affricate is a good vehicle for ç, but ö is unwritable and a second syllable is added. 牙 <i>ɲa</i> → <i>ok</i> : a velar-nasal initial standing for a vowel-initial syllable is unsupported. REASON <i>çöge</i> of height or eminence, with the same <i>ok</i> element.				
烏珠留若鞮 Wuzhuliu Ruodi	ʔa tso liu ɳak te	’a ço liu nyak te	Çal Inakt	the grey or dappled one, of the Inakt
PHONETICS 烏 <i>ʔa</i> is vowel-initial, so the reading begins at 珠 <i>tso</i> → ç, with 留 <i>liu</i> → l: an affricate written with the default tʃ- character and a liquid, both majority correspondences. Nothing is inserted, and no character is read as a consonant it does not carry. REASON <i>çal</i> is “alaca, kır” in Kâşgarî — dappled, grey. Colour epithets are a persistent Turkic naming pattern, as <i>çakır</i> “grey-eyed” shows in the same dictionary.				
烏累若鞮 Wulei Ruodi	ʔa lui ɳak te	’a lui nyak te	Uluğ Inakt	the Great, of the Inakt
PHONETICS 烏 <i>ʔa</i> → <i>u</i> : a glottal-onset character is the standard way to write a vowel-initial syllable. 累 <i>lui</i> → <i>lu</i> : a liquid, direct. Two characters, two segments, nothing supplied that the transcription does not carry. REASON The Codex Cumanicus glosses <i>Oulu</i> as <i>magnus</i> . <i>Uluğ</i> “great” is among the most common elements in Turkic titlature, which is both the strength of this reading and its weakness: a very common word is an easy target to hit.				
呼都而尸道皋若鞮 Huduershidaogao Ruodi	ha ta ɳə ʃi dou ^B kou ɳak te	ha ta nyı ʃi dou kou nyak te	Öz-Uluğ Inakt	Great Core, of the Inakt
PHONETICS Eight characters, and correspondingly the least constrained reading in the file: the longer the string, the more ways there are to split it into Turkic elements, and the split itself is doing most of the work rather than any individual correspondence. 呼 <i>ha</i> → <i>öz</i> and 皋 <i>kou</i> → <i>uluğ</i> are the two largest jumps. REASON <i>öz</i> “self, core” and <i>uluğ</i> “great”, both common Old Turkic elements in titlature.				

TABLE S3 Six later chanyu whose formal names end in 若鞮. Only the shared element is argued for in the paper; the material preceding it in each name is a proposal like any other in this file.

S1.8 The ruling clan name, in its two written forms

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING PROPOSED BY THE AUTHOR	SENSE PROPOSED FOR THAT READING
攣鞮 Luandi	lyan te	lyan te	El-indi	state founder
PHONETICS 攣 <i>lyan</i> → <i>el</i> moves the liquid from the onset of the Chinese syllable to the coda of the Turkic one; no corpus shows that kind of metathesis. 鞮 <i>te</i> → <i>indi</i> expands one syllable into two. The reading is driven by the sense rather than by the transcription. REASON <i>el</i> “realm, state” with the definite past in <i>-di</i> : literally “he founded the realm”. This is the clan name as the <i>Shiji</i> and <i>Hanshu</i> write it.				
虛連題 Xulianti	kʰia lian ^B de	kia lian de	Kıyalığ El-indi	state founder of the rocky region
PHONETICS 虛 <i>kʰia</i> → <i>kıya</i> is reachable. 連 <i>lian^B</i> → <i>lığ</i> adds an unwritten coda, and 題 <i>de</i> → <i>el-indi</i> carries a whole two-syllable word on one character. Three characters are being asked to yield five syllables. REASON The same clan name as 攣鞮, in the form the <i>Hou Hanshu</i> gives, with the <i>kaya</i> element recurring from 虛閭權渠.				

TABLE S4 攣鞮 in the *Shiji* and *Hanshu*, 虛連題 in the *Hou Hanshu*. Both are taken here as the same name, which is the usual view.

S1.9 Titles and lexical items

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING PROPOSED BY THE AUTHOR	SENSE PROPOSED FOR THAT READING
單于 chanyu	džan wa	can va	Kan Apa	senior ruler, elder ruler
PHONETICS 單 <i>džan</i> → <i>kan</i> is the weak step: a voiced affricate standing for a velar stop is not a correspondence any of the three corpora show. 于 <i>wa</i> → <i>apa</i> expands one syllable into two. This is one of the five items decoded entirely from character-specific evidence, which makes the mismatch harder to attribute to sparse data. REASON <i>apa</i> “elder, ancestor” is attested in Old Turkic titlature. This is the central title of the whole record and the most contested: readings from three language families have been proposed for it.				
撐犁 chengli	ɢaŋ lei	dang lei	Tanglı	bearing divine light; sacred
PHONETICS 撐 <i>ɢaŋ</i> → <i>tang</i> : retroflex ɢ for t is routine and the -ŋ coda is written. 犁 <i>lei</i> → <i>lı</i> : direct. Both steps are default. The same two syllables equally support <i>tengri</i> , which the Chinese gloss 天 and the lexicon search of §6.3 both favour — Kâşğarî returns it 4th, the Codex 2nd. REASON <i>tang</i> of dawn or light, read as “bearing divine light; sacred”. Offered as an alternative to the standard identification with <i>tengri</i> , not as an improvement on it.				
當戶 danghu	taŋ ga ^B	tang ga	Tanga	Siberian tiger or leopard
PHONETICS 當 <i>taŋ</i> → <i>tan</i> : direct, with the -ŋ coda written and simply read as -n. 戶 <i>ga</i> → <i>ga</i> : direct. Phonetically the least demanding reading in the file after 服匿 — nothing is inserted and nothing unwritable is required. REASON <i>tonga</i> “Siberian tiger, leopard”, the word in the name of the hero of the Alp Er Tunga epic. Big-cat epithets are a persistent Turkic pattern: Alp Arslan (<i>arslan</i> , lion), Baybars (<i>bars</i> , a large cat).				
閼氏 yanzhi / ezhi	ʔat dže ^B	’at ce	Ece	queen
PHONETICS 閼 <i>ʔat</i> → <i>e</i> : a vowel-initial character, but with a -t coda that is written and that the reading discards. 氏 <i>dže^B</i> → <i>ce</i> : direct. Another of the five fully character-anchored items, so the discarded -t is a real cost rather than an artefact. REASON <i>ece</i> “queen”, which survives in modern Turkish with roughly this sense. 閼氏 is the title of the chanyu's principal consort.				
骨都侯 guduhou	kuət ta go	kuit ta go	Kutak	the blessed one
PHONETICS 骨 <i>kuət</i> → <i>kut</i> : default velar onset with the -t coda written — the tightest single syllable in this file. 都 <i>ta</i> then begins with the same dental, and step 22 measures what that means: at a junction where a written coda is followed by the same-place onset, the source word carries that consonant once rather than twice in 121 of 140 cases (86%). So 骨都 transcribes <i>kuta</i> , not <i>kutta</i> . 侯 <i>go</i> → <i>-k</i> closes it, a Later Han voiced g- standing for a source k more often than for a source g. REASON <i>kut</i> is “kut, uğur, devlet, baht, talih, saadet” in Kâşğarî — heavenly favour, fortune, sovereign grace — and Clauson records the verb <i>kutad</i> - “to be blessed”, the stem behind the <i>Kutadgu Bilig</i> . <i>Kutak</i> itself is not attested : it would be a deverbal or diminutive formation in -(A)k on that stem, which is a live suffix but not one witnessed on this word. Compare Murad I's <i>Hüdavendigâr</i> and <i>Qutalmış</i> , ancestor of the Seljuks of Rum. Note also that <i>kut</i> is the Turkic candidate for 孤塗, where the Chinese gloss 子 “son” fits the Yeniseian proposal better and not this one.				
且渠 juqu	tsa ^B ɢia	tsa ɢia	Çaka	the striker; from çak- ‘to strike fire, to strike a blow’
PHONETICS 且 <i>tsa^B</i> → <i>ça</i> : ts- for a source affricate is a minority spelling but an attested one — 15% in the Sanskrit corpus, 9% in Ligeti — against a ts̥-character as the default at 37%. 渠 <i>ɢia</i> → <i>ka</i> : a Later Han voiced g- stands for a foreign k more often than for a foreign g. Both steps are permitted by the measured correspondences. REASON <i>çak</i> - “to strike fire from flint, to sting, to slander”, which Clauson describes as onomatopoeic in character and of violent action, physical or mental. No <i>çaka</i> headword exists in Kâşğarî, the Codex or Clauson — only the verb and its derivatives <i>çakmak</i> , <i>çakır</i> , <i>çakılmak</i> — so <i>Çaka</i> would have to be the -A converb of that stem, and a bare converb serving as an office title is not a demonstrable pattern. The 11th-century emir Çaka Bey of Smyrna (Byzantine Τζαχάç, fl. 1081–93) shows that <i>çak</i> - could yield a personal name, but that is a personal name set against a title. The Chinese sources do not record what the office of 且渠 did, so no sense can be tested against a gloss.				
徑路 jinglu	keŋ ^c la ^c	keng la	Kingrak	a large cleaver-like knife

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING PROPOSED BY THE AUTHOR	SENSE PROPOSED FOR THAT READING
PHONETICS 徑 <i>keŋ</i> → <i>king</i> : default velar onset with the -ŋ coda written. 路 <i>la</i> → <i>ra</i> : a Chinese l- for a source r-, which step 15 shows is the ordinary direction — a source r- was written l- in 12 of 14 attestations. The final -k of the Turkic word is not written, and step 20 licenses that: 30% of stop-final source syllables were spelled with an open character. REASON This reading is not the author's. The comparison of 徑路 with Turkic <i>kıñrak</i> is already in the literature, where it is described as speculative; it entered this project during the corpus build, recorded as cited and unverified. What is new here is that it can be tested, and it survives. Kāšgarī glosses <i>kıñrak</i> as “et ve hamur kesilen satıra benzer büyük bıçak” — a large knife resembling a cleaver — and the Chinese sources independently describe 徑路 as a Xiongnu blade used in oath-taking (the 徑路刀). Two sources with no contact agree about what the object is, which is a stronger kind of agreement than a sound match. A word of identical shape, <i>kañrak</i> “palate”, fits the transcription equally well and is excluded only by the referent. §9.1 of the paper sets this out; the original publication of the comparison still has to be traced.				
服匿 funi	buk ɲik	buk nık	Buknik	clay jar (küp)
PHONETICS 服 <i>buk</i> → <i>buk</i> : exact, including the written -k coda. 匿 <i>ɲik</i> → <i>nik</i> : retroflex ɲ for n is routine and the second -k is also written. This is the closest fit to its own transcription of anything in this file — no segment inserted, none discarded. REASON A clay jar (Turkish <i>küp</i>). One of the few Xiongnu material terms the Chinese sources preserve, which is what makes it worth a reading at all.				
甌脫 outuo	ʔo tʰuat	’o tuat	Otuyat	the border watchtowers
PHONETICS 甌 <i>ʔo</i> → <i>o</i> : vowel-initial, direct. 脫 <i>tʰuat</i> → <i>tuyat</i> : the aspirated t- is direct and the -t coda is written; the only addition is the medial -y-. REASON The border watchtowers, or the empty zone between Xiongnu and Han territory, for which the Chinese sources use 甌脫.				
胡祿 hulu	go lok	go lok	Külük	glorious
PHONETICS 胡 <i>go</i> → <i>kü</i> : the devoicing is normal, since a Later Han voiced g- stands for a foreign k more often than for a foreign g, but ü cannot be written and surfaces as o. 祿 <i>lok</i> → <i>lük</i> : the -k coda is written. The consonant frame k-l-k is secure; only the vowels are free. REASON <i>külüg</i> “famous, renowned”, an Old Turkic epithet of praise.				

TABLE S5 The titles and the handful of common nouns the Chinese histories preserve. The lexical items are the more interesting group because some of them are glossed in the source, which means a proposal about them can in principle be checked.

S1.10 Ethnonyms

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING PROPOSED BY THE AUTHOR	SENSE PROPOSED FOR THAT READING
匈奴 Xiongnu	huoŋ na	huong na	Hung	the ethnonym, from the first character alone
PHONETICS 匈 <i>huoŋ</i> → <i>hung</i> : direct, with the -ŋ coda written. 奴 <i>na</i> is discarded rather than read, on the assumption that 奴 “slave” is a Chinese pejorative appended to the name rather than part of it. REASON Dropping 奴 is a common move in the literature and is not original here; what is proposed is only the vocalisation of the remaining syllable.				
羯 Jie	kɪat	kɪat	Kiyat	floods
PHONETICS A single character. 羯 <i>kɪat</i> → <i>kiyat</i> : the k- is default and the -t coda is written, so the consonant frame k...t is secure and only the medial glide is free. Fully character-anchored. REASON <i>kiyat</i> , of flooding. The Jie are recorded as a group within or beside the Xiongnu; the reading is offered for the ethnonym only.				

TABLE S6 Group names rather than personal names, and subject to a further difficulty: a Chinese exonym need not transcribe anything the group called itself.

S1.11 How to disagree

The repository includes `src/lookup_lexicons.py`, which searches Kāšgarī’s *Dīwān*, the *Codex Cumanicus* and any EPUB lexicon placed beside them, folding the spelling conventions that separate Turcological from Turkish orthography (q ≡ k, ñ ≡ ng, ş ≡ s, ı ≡ i). Type any proposed form into it and see what the medieval sources actually have. That is how the *Codex* attestation for *inak* was found, and how the -tI count that ruled out the competing suffix was made.

A reader who wants to dispose of a row below should find that easier than the author did. The useful form of disagreement is not “this is unlikely” — most of them are — but “this word is not attested before date X”, or “this suffix does not attach to that stem”, or “the Chinese

character cannot carry that coda". Those are the objections that would move something out of this file, in either direction.

S1.12 Status

Every reading in Tables S2–S6 carries the same label in the machine-readable file: *author's proposal; not supported by any measurement in the paper*. That label is the honest summary of all thirty-three, and it is not weakened by the fact that some of them are more attractive than others. The one reading with a different status, 若鞮 → *Inakt*, is argued in §9 of the paper and summarised in §S1.4 above.

S1.13 Sources referred to in this supplement

1. Clauson, G. (1972). *An Etymological Dictionary of Pre-Thirteenth-Century Turkish*. Oxford: Clarendon Press.
2. Erdal, M. (2004). *A Grammar of Old Turkic*. Leiden: Brill.
3. Golden, P. B. (1992). *An Introduction to the History of the Turkic Peoples*. Wiesbaden: Harrassowitz.
4. Kâşgarlı Mahmud (1072–77). *Dīwān Lughāt al-Turk*. Index volume (*Dizin*), Türk Dil Kurumu Yayınları.
5. Kuun, G., ed. (1880). *Codex Cumanicus Bibliothecæ ad Templum Divi Marci Venetiarum*. Budapest: Editio Scient. Academiæ Hung.
6. Schuessler, A. (2009). *Minimal Old Chinese and Later Han Chinese*. Honolulu: University of Hawai'i Press.
7. Uluğ, A. M. (2026). *chinese-transcription-channel* [software and data]. github.com/ulug2004/chinese-transcription-channel